BirdViz

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Data Visualization Tool for Bird Sightings in US Ecological Centers

Goal: Develop a tool to explore and interpret bird sightings data

Data Sources: Macaulay Library and National Ecological Observatory Network (NEON) breeding birds dataset

Key Features:

- Maps and graphical visualization
- Identification analysis
- Species distribution visualization

Significance: Addresses the challenge of deriving meaningful insights from vast ecological datasets in the digital age

Datasets

- **Macaulay Library**
  - Nearly 2.4 million sound recordings of birds
  - Extensive collection of bird images and videos
  - Detailed information on bird sightings
  - Metadata associated with each unique bird species

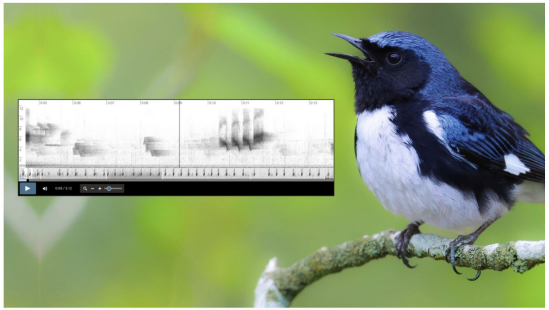


Image: Black-Throated Blue Warbler © Ryan Justice, Macaulay Library / Recording: Black-Throated Blue Warbler © Bob McGuire, Macaulay Library

A Barred Owl (Strix varia) at the Disney Wilderness Preserve field site in Florida.

- **National Ecological Observatory Network (NEON) breeding birds dataset**
  - Bird count information collected by experienced observers
  - Data from 5-15 sampling grids at most NEON field sites
  - Observations made during the peak breeding season for resident birds
  - Records of species, sex, and distance from observation points for each bird sighted or heard

Software requirements

- **Python**: create the graphical representations
- **PyDeck**: Render maps and display information on maps
- **HTML**: Create the web interface for displaying all the graphical representations.
- **Javascript**: Enable interaction with the user interface

Challenges Encountered

Feature selection

- Large number of available CSV files, each having about 25 features
- Difficulty in identifying most relevant attributes
- Balancing between comprehensive data and focused analysis

Handling missing data

- Inconsistent observations across sites and years
- Handled by removing those sites and years

Creating a unified interface

- Integration of individual visualizations
- Development of coherent user experience
- Combining interactive map and graphs
- Ensuring proper responsiveness as python does not have good means of interactiveness

Moving to demo our Project!

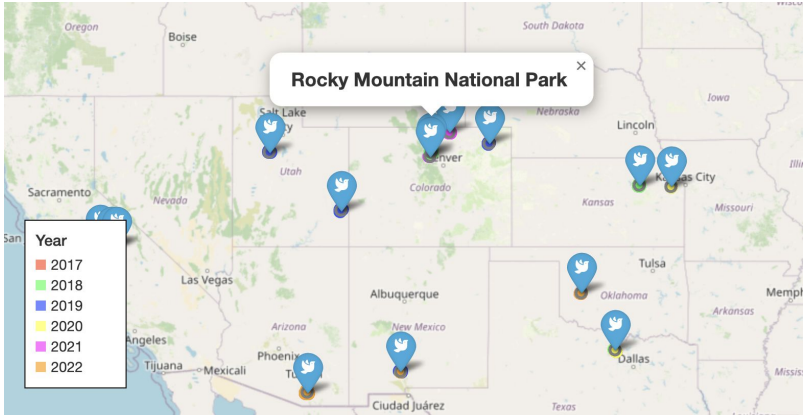
Visualization - Interactive Map

An interactive map to display observation points, color-coded by year.

Vis channels and encoding:

- Position: Latitude and longitude of observations
- Color: Year of observation
- Shape: Circle markers for observations, larger circles for site locations
- Icon: Bird icon for site markers

Justification: The map provides a spatial overview of observation locations. Color-coding by year allows users to see temporal patterns in the spatial distribution. The use of different shapes and sizes for observations vs. sites helps distinguish between them.



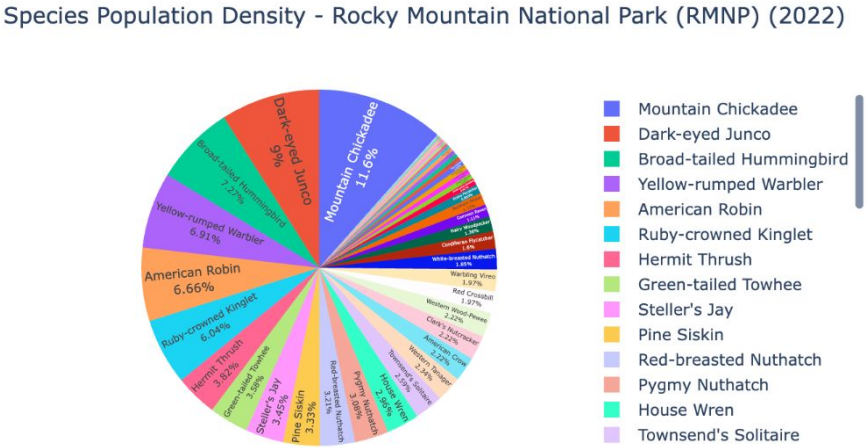
Visualization - Species Population Density

The Species Population Density pie chart visualizes the relative abundance of different bird species observed at a selected site for a **specific year**.

Vis channels and encoding:

- Angle/Area: Proportion of each species
- Color: Different color for each species
- Labels: Common names of species

Justification: The use of a pie chart for this data is appropriate as it clearly shows the composition of the bird population, allowing for easy comparison between species within a single year and location.



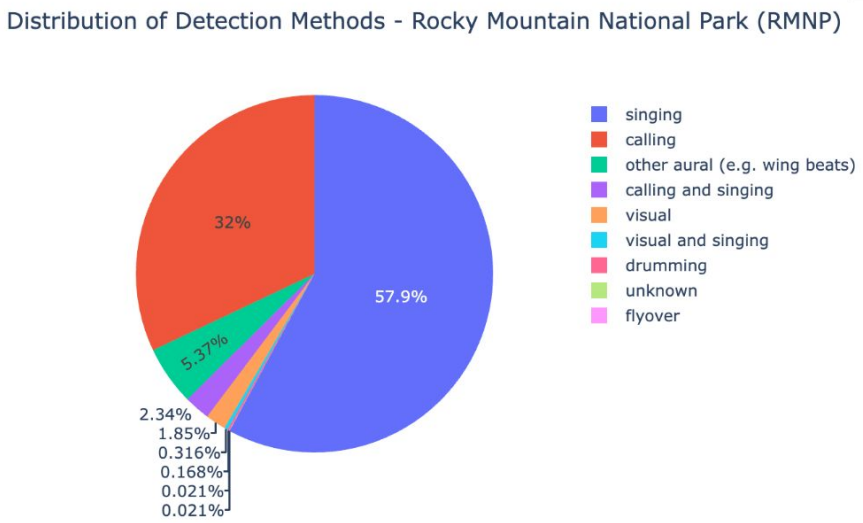
Visualization - Distribution of Detection Methods

The detection method pie chart visualizes the distribution of different methods used to detect bird species at the selected site.

Vis channels and encoding:

- Angle/Area: Represents the proportion of each detection method
- Color: Different colors for each detection method
- Labels: Names of detection methods

Justification: A pie chart is effective for showing the relative proportions of different categories within a whole. In this case, it clearly illustrates the prevalence of various detection methods used in bird observations. The use of distinct colors for each method helps in quickly distinguishing between different techniques.



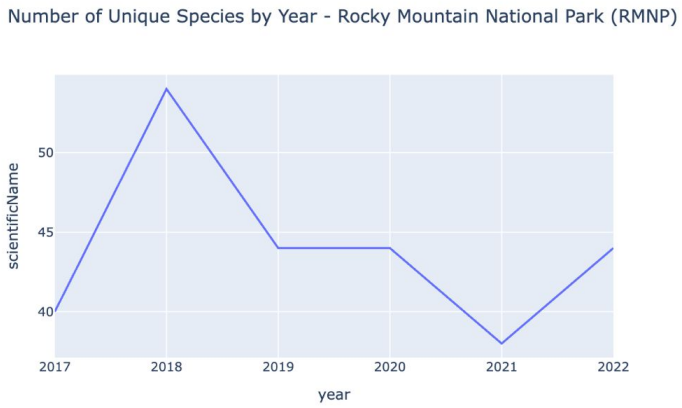
Visualization - Number of Unique Species by Year

A line graph to show how the number of unique species changes over time.

Vis channels and encoding:

- X-axis: Year
- Y-axis: Number of unique species
- Line: Connects data points
- Points: Marks each year's data point

Justification: Line graphs are excellent for showing trends over time. The addition of points helps highlight specific yearly values.



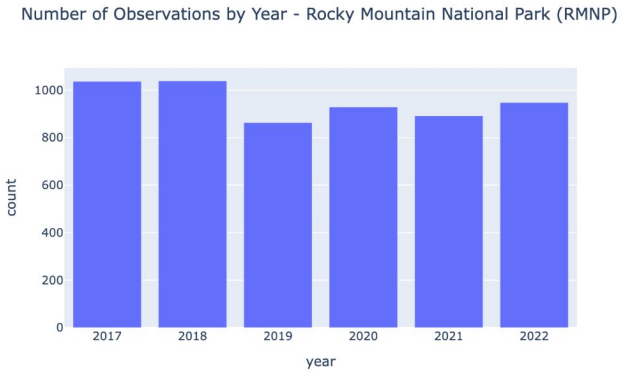
Visualization - Number of Observations by Year

A bar chart to show the count of observations per year.

Vis channels and encoding:

- X-axis: Year
- Y-axis: Count of observations
- Color: Light blue bars

Justification: Bar charts are effective for comparing quantities across categories (years in this case). The use of a single color (light blue) keeps the chart simple and easy to read.



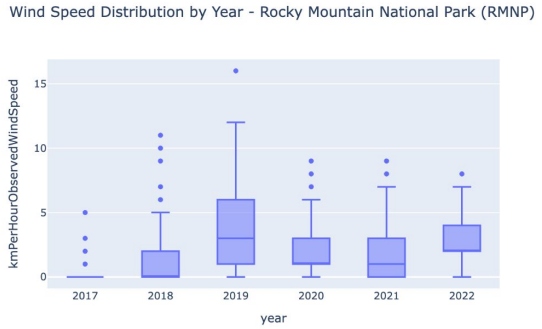
Visualization - environmental conditions by Year

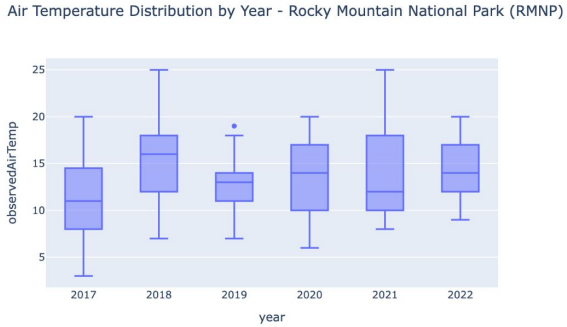
A box plot to show the distribution of wind speed or air temperatures for each year.

Vis channels and encoding:

- X-axis: Year
- Y-axis: Observed wind speed (kmph) or air temperature (celcius)
- Box and whiskers: Distribution of wind speed or air temperatures

Justification: Box plots are ideal for showing distributions and allowing comparisons across categories (years). They effectively display median, quartiles, and outliers.





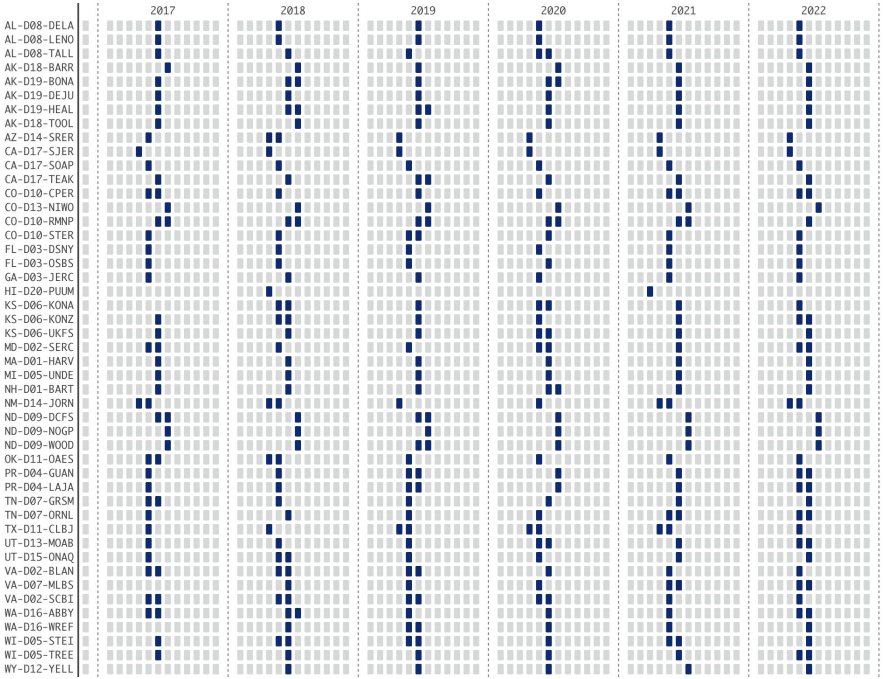
Evaluation

How efficiently the data is represented?

- Create a tool that allows users to understand data
- Use graphical representations
- Handle large amounts of tabular and numerical data.
- Provide a unified interface

NEON Visualization

1. Only temporal records
2. No proper site-name mapping
3. No fine grained visualizations about geolocations of sighting
4. No visualization to summarize species density, population changes, and species counts



Detailed Analytical Questions:

- How do bird species distributions vary spatially and temporally across different ecological centers?
- What are the predominant methods used for bird sightings, and what do they reveal about bird behavior in relation to human presence?
- What patterns emerge in bird density and species diversity among various ecological centers?
- How does the distribution of bird species change over time in specific locations?

Q. How do bird species distributions vary spatially and temporally across different ecological centers?

Achieved meaningful summarization of spatio-temporal attributes of bird sightings through an interactive map that allows users to explore geolocations of bird sightings across different NEON sites.

Q. What are the predominant methods used for bird sightings, and what do they reveal about bird behavior in relation to human presence?

The interface includes meaningful graphics describing population density, environmental factors, and detection methods over time.

Q. What patterns emerge in bird density and species diversity among various ecological centers?

Visualized population density and environmental factors over time to help uncover patterns in species density and diversity across ecological centers.

Q. How does the distribution of bird species change over time in specific locations?

Captured temporal attributes of bird sightings allowing temporal analysis of species distributions in specific locations.

Summary of Achievements and Updates to the Proposal

- **Achievements:**
  a. Developed an interface that displays the geolocation of bird sightings recorded in different NEON sites
  b. Managed to meaningfully summarize the spatio-temporal attributes of bird sightings
  c. Interactive map that allows users to pan and zoom over different clearly delineated geo-locations
  d. Display meaningful graphics to describe the population density, environmental factors, and detection methods over time.
- **Scope for improvement:**
  a. Mostly able to achieve the initial planned framework.
  b. Create interactions between the map and the statistics.
  c. Update graphs in real-time

Thank you!